

# Failure budgets and service objectives

Project: Grounded Learning Service

## 1. Prepare the purpose, prerequisites and materials

### Teach the learning cycle

Primary target: Explain the mechanism and terminology of failure budgets and service objectives

#### Session · Sequence and minutes · Resources

**Session:** Session 1 · Understand · 45 min

**Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.

**Resources:** Sections 1–6; record the brief recall in the workbook.

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**Session:** Session 2 · Practise · 45 min

**Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

**Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

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**Session:** Session 3 · Investigate and transfer · 45 min

**Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.

**Resources:** Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

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This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building. Repeated copies of a case share one recorded attempt; they are not new evidence.

## Retrieval answer guidance

### Evaluation pipelines

Without opening the old lesson, explain one key idea from Evaluation pipelines. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the mechanism and terminology of evaluation pipelines

Reference example, not the only acceptable wording: Teacher explanation: Evaluation pipelines preserve case versions, procedure versions and raw outcomes before producing metrics. A rerun must identify what changed. Load versioned cases, validate expected outcomes, execute the frozen system, store case-level results and compute declared denominators. Reject missing results rather than silently counting them as success.

### Testing model-dependent behaviour

Without opening the old lesson, explain one key idea from Testing model-dependent behaviour. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the mechanism and terminology of testing model-dependent behaviour

Reference example, not the only acceptable wording: Teacher explanation: Model-dependent behaviour may vary between runs. Evaluate defined semantic criteria and report repeated outcomes rather than requiring one exact sentence. Use fixed evidence and rubric per case. Require supported room/capacity, valid schema and no actions. Repeats estimate observed variation for this case; they are not additional independent users.

### Authentication and authorisation

Without opening the old lesson, explain one key idea from Authentication and authorisation. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the mechanism and terminology of authentication and authorisation

Reference example, not the only acceptable wording: Teacher explanation: Authentication identifies an actor; authorisation decides whether that actor may perform a particular operation on a particular resource. Use trusted session roles, never a role claimed inside user JSON. Student may read student resources; teacher may read both. Unauthenticated requests are denied.

Two 45-minute sessions. Service objectives define measured reliability over a stated interval. A failure budget is meaningful only after success and eligible requests are defined. Use the supplied constructed scenarios to make the mechanism inspectable. Collect W4 before sharing solutions. Students should name intermediate evidence and limitations; classroom pacing and learning effectiveness remain unvalidated.

### Outcomes

Explain the mechanism and terminology of failure budgets and service objectives

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

## Preparation

Lesson and workbook; paper and pencil

Calculator and optional local Python 3 for arithmetic; no live API or private data

Prepare the fresh retry separately and withhold its solution until after collection

## Prerequisite answer

A result depends on the inputs, procedure and conditions that produced it. Preserve data roles and distinguish observed outcomes from predictions or unmeasured conditions. If the prior concept cannot be explained, use a short retrieval activity before this chapter.

## End goal for the pictorial case

Explain failure budgets and service objectives using the deciding step in this fictional service.

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

# 2. Explain and model the mechanism

Service objectives define measured reliability over a stated interval. A failure budget is meaningful only after success and eligible requests are defined.

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing. These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

For a fictional objective at least 98% of 200 eligible requests must pass all contract checks. Allowed failures are  $2\% \cdot 200 = 4$ . Count every eligible failure once; report excluded requests separately.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same inputs. If a required input is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

### Term · Meaning

**Term:** Service objective

**Meaning:** A declared measured target over a specified scope

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**Term:** Failure budget

**Meaning:** The failures allowed under that target

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**Term:** Eligible request

**Meaning:** A request included under the declared denominator rule

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## Failure budgets and service objectives — evidence trace

**Input and scope · Check or calculation · Conclusion and limit**

**Input and scope:** In the interval 200 requests are eligible and 5 fail.

**Check or calculation:** For a fictional objective at least 98% of 200 eligible requests must pass all contract checks.

**Conclusion and limit:**  $195/200=97.5\%$ , below 98%; failures exceed budget 4 by one. This is the measured interval only.

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The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Guided:  $195/200=97.5\%$ , below 98%; failures exceed budget 4 by one. This is the measured interval only.  
Qualification: Meeting one interval objective does not establish every future interval or zero critical failures.  
Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the mechanism; reject a claim resting on unmeasured evidence.

## Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a prediction, then explain the deciding step. These teacher notes contain solutions; do not reveal the independent case answer during modelling.

In the interval 200 requests are eligible and 5 fail.

$195/200=97.5\%$ , below 98%; failures exceed budget 4 by one. This is the measured interval only.

Compare the intermediate evidence with the final statement. Explain why each number or decision follows from the stated inputs. The conclusion belongs to this scenario and procedure; changing the inputs, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

**Thinking move · Teacher prompt**

**Thinking move:** Plan

**Teacher prompt:** What is given? What is the question? Which procedure fits these facts?

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**Thinking move:** Do

**Teacher prompt:** Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.

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**Thinking move:** Check

**Teacher prompt:** Does the output answer the question? Which assumption or limitation prevents a larger claim?

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Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

### A pictorial worked case: Failure budgets and service objectives

A reliability allowance. A practice service requires at least 98 successes in each window of 100 eligible requests.

#### The facts we will use

Current window has 97 successes and 3 failures. Success includes a correct refusal within the agreed time limit.

AI ACADEMY / AI-6/11

### Failure count in one 100-request window

failed requests out of 100

Allowed maximum



Observed



Bar length shows the supplied value; the marked baseline is zero.

## Failure budgets and service objectives

### Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



#### 1. Define success

Use the same predeclared correctness and timing criteria.



#### 2. Compute allowance

The target permits at most 2 failures in 100 requests.



#### 3. Compare record

Three failures misses the target by one request.

### CHANGE ONE THING

Repair one failed request into a success in this window.

Explain the changed result using the matching lesson.

### Read the picture one step at a time

#### Step · What happens and why

**Step:** 1. Define success

**What happens and why:** Use the same predeclared correctness and timing criteria.

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**Step:** 2. Compute allowance

**What happens and why:** The target permits at most 2 failures in 100 requests.

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### Step: 3. Compare record

**What happens and why:** Three failures misses the target by one request.

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Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

### Why this result follows

An objective needs a measurement window, denominator and success definition. An honest refusal can be success under an agreed contract.

### What this example does not establish

Meeting one window does not guarantee the next; monitor repeated windows and distinguish failure types.

### Change one thing in the pictured case

Change: Repair one failed request into a success in this window.

Prediction: 98 successes and 2 failures meet the stated window target.

Reason: One recorded outcome changes while the denominator stays 100.

### Ask while showing the picture

Cover the result. Ask learners to name the inputs and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its evidence. Then change the supplied condition and compare. The picture is a representation; it does not document execution of a real AI model.

## 3. Demonstrate and guide practice

### Time or stage · Teacher action

**Time or stage:** Session 1: 0–7 minutes

**Teacher action:** Retrieve the prerequisite; identify the input, procedure and evidence source.

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**Time or stage:** 7–18 minutes

**Teacher action:** Explain section 1 and vocabulary. Complete W1 with a mechanism-based explanation.

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**Time or stage:** 18–32 minutes

**Teacher action:** Trace the declared procedure and worked case. Complete W2, recording intermediate steps.

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**Time or stage:** 32–45 minutes

**Teacher action:** Discuss the diagram with the worked input; use W3 to distinguish measured claims and limitations.

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**Time or stage:** Session 2: 0–8 minutes

**Teacher action:** Retrieve the worked reasoning and collect the guided response before discussing it.

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**Time or stage:** 8–20 minutes

**Teacher action:** Compare a proposed boundary variant with the original case; identify what remains fixed.

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**Time or stage:** 20–35 minutes

**Teacher action:** Collect W4 independently. Require the procedure trace and the bounded conclusion.

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**Time or stage:** 35–45 minutes

**Teacher action:** Use W5 and exit responses for feedback. Administer the fresh retry after feedback and preserve both attempts.

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Use the worked example in Sections 3 and 6. Collect a prediction before revealing its result; ask the learner to point to each intermediate fact.

Provide an input/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace. Extension: compare two concrete variants of this mechanism while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

## Reduce help deliberately

Completion task: In the interval 200 requests are eligible and 5 fail. Show the intermediate steps and give the conclusion with its scope.

Teacher check: Teacher explanation:  $195/200=97.5\%$ , below 98%; failures exceed budget 4 by one. This is the measured interval only.

Start by identifying the given inputs together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood. For partners, alternate explainer and checker; collect a response from every learner before discussion.

## Individual reasoning check

Use the worked case for Failure budgets and service objectives. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: Meeting one interval objective does not establish every future interval or zero critical failures.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class. If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

### **Pictorial practice question**

Calculate the maximum allowed failures among 100 requests. Does 97 successes and 3 failures meet the target, and what changes after one failure becomes a success?

### **Reasoned teacher answer**

Use the same predeclared correctness and timing criteria. The target permits at most 2 failures in 100 requests. Three failures misses the target by one request. Changed case: 98 successes and 2 failures meet the stated window target. One recorded outcome changes while the denominator stays 100.

Changed-case reasoning: 98 successes and 2 failures meet the stated window target. One recorded outcome changes while the denominator stays 100.

Collect each learner’s explanation before discussion. Use the first missing connection between input, deciding step and result to choose support. This question is worked-case practice, not fresh mastery evidence.

## **4. Run the investigation and diagnose misconceptions**

The worked result establishes every future case: Meeting one interval objective does not establish every future interval or zero critical failures.

A missing input may be replaced without recording the change: A replacement changes the evidence or contract. Record the change and its justification; do not silently turn unknown into a measured value.

Repeating an assessed case is new independent evidence: A replay checks consistency. An example used for feedback or revision cannot later be described as untouched final evaluation.

Use Section 9 for the evidence ledger. Add a boundary or missing-input case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

### **Match the support to the first error**

#### **What the response shows · Next teaching action**

**What the response shows:** Missing or misread facts

**Next teaching action:** Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.

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**What the response shows:** Wrong procedure

**Next teaching action:** Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.

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**What the response shows:** Arithmetic or implementation error

**Next teaching action:** Trace intermediate values or lines; preserve the first mismatch and retry that step.

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**What the response shows:** Correct result without a reason

**Next teaching action:** Ask for a trace and a changed case before marking independent understanding.

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**What the response shows:** An unsupported wider claim

**Next teaching action:** Ask which supplied evidence supports the claim and which further observation would be needed.

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Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal. Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

## 5. Assess and explain the answers

### Guided and exit questions

Guided:  $195/200=97.5\%$ , below 98%; failures exceed budget 4 by one. This is the measured interval only. Qualification: Meeting one interval objective does not establish every future interval or zero critical failures. Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the mechanism; reject a claim resting on unmeasured evidence.

### W1 Explain the mechanism

Explain the weekly mechanism in your own words. Use the worked scenario to name its input, decision rule and output.

Teacher explanation: Service objectives define measured reliability over a stated interval. A failure budget is meaningful only after success and eligible requests are defined. For a fictional objective at least 98% of 200 eligible requests must pass all contract checks. Allowed failures are  $2\% \cdot 200 = 4$ . Count every eligible

failure once; report excluded requests separately.

## W2 Trace the worked case

In the interval 200 requests are eligible and 5 fail. Show the intermediate steps and give the conclusion with its scope.

Teacher explanation:  $195/200=97.5\%$ , below 98%; failures exceed budget 4 by one. This is the measured interval only.

## W3 Audit the limit

A report omits this qualification: Meeting one interval objective does not establish every future interval or zero critical failures. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Teacher explanation: Meeting one interval objective does not establish every future interval or zero critical failures. State the observed result separately from the unverified or out-of-scope claim; preserve the original evidence rather than inventing a missing fact.

## W4 Apply a new case independently

Objective 95% over 80 eligible requests. Five fail. Compute allowed failures and observed success.

Teacher explanation: Allowed four;  $75/80=93.75\%$ , so objective fails. Unknown eligibility cannot be silently excluded.

## W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison. Do not reuse W4 as a new final test after receiving feedback.

Teacher explanation: Accept a concrete variant of the worked input or a missing/conflicting/boundary condition, with an expectation derived from the declared procedure. Require preserved settings, original outputs and data roles. Meeting one interval objective does not establish every future interval or zero critical failures. The new case should test this particular mechanism rather than merely restate the original result.

## Independent assessment and fresh retry

W4: Allowed four;  $75/80=93.75\%$ , so objective fails. Unknown eligibility cannot be silently excluded. Require intermediate reasoning and the stated scope. Fresh retry: Objective 90% over 50 requests with four failures. Expected reasoning: Allowed five;  $46/50=92\%$ , so interval meets objective. If students need support, record it and reassess with another new case rather than awarding independence to a copied answer.

## Assess independent transfer

Selected case: Objective 95% over 80 eligible requests. Five fail. Compute allowed failures and observed success.

## Selected case answer

Teacher explanation: Allowed four;  $75/80=93.75\%$ , so objective fails. Unknown eligibility cannot be silently excluded.

### Dimension · Evidence required

**Dimension:** Explanation

**Evidence required:** Names the applicable idea or procedure and ties it to the supplied facts.

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**Dimension:** Trace and result

**Evidence required:** Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.

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**Dimension:** Limits

**Evidence required:** States the assumptions, missing evidence or conditions under which the result would change.

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Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace. Do not penalise an honestly recorded model failure if the learner correctly explains its evidence and limits.

## Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case. Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

## 6. Connect the project, support and next lesson

Add a labelled failure budgets and service objectives evidence sheet to Grounded Learning Service. Show how you can explain the mechanism and terminology of failure budgets and service objectives. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Provide an input/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace. Extension: compare two concrete variants of this mechanism

while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

Next connection: Logs traces and privacy. Preserve the current case record and retrieve its evidence boundary before moving on.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

## Plan the delayed return

Suggested return points in this level: ai-6/12, ai-6/14, ai-6/17. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used. For a new level, revisit the previous capstone before checking the new prerequisites.

### Record · Teacher evidence

**Record:** Learner code and date

**Teacher evidence:** \_\_\_\_\_

\_\_\_\_\_

**Record:** Concept and case used; prior exposure

**Teacher evidence:** \_\_\_\_\_

\_\_\_\_\_

**Record:** Explanation / trace / limits

**Teacher evidence:** \_\_\_\_\_

\_\_\_\_\_

**Record:** Support and first repair

**Teacher evidence:** \_\_\_\_\_

\_\_\_\_\_

**Record:** Changed case and delayed outcome

**Teacher evidence:** \_\_\_\_\_

\_\_\_\_\_

**Record:** Next teaching action

**Teacher evidence:** \_\_\_\_\_

\_\_\_\_\_

Use the evidence to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

### **Project evidence from this case**

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or evidence that makes the deciding step valid, so another team can review it.